

1) Given a network address of 192.168.10.0/24, divide it into 4 equal subnets and write down the subnet mask, network address, and broadcast address for each subnet.

Step 1: Determine the New Prefix

- Original network: 192.168.10.0/24.
- Four equal subnets require borrowing 2 host bits because $2^2 = 4$.
- New prefix: /26.

Step 2: Calculate the Subnet Mask

- The /26 subnet mask is 255.255.255.192.
- The block size is $256 - 192 = 64$ addresses per subnet.
- Each /26 subnet has 64 total addresses and 62 usable host addresses.

Step 3: Subnet Address Table

- Subnet 1 — Network: 192.168.10.0/26 — Usable: 192.168.10.1–192.168.10.62 — Broadcast: 192.168.10.63.
- Subnet 2 — Network: 192.168.10.64/26 — Usable: 192.168.10.65–192.168.10.126 — Broadcast: 192.168.10.127.
- Subnet 3 — Network: 192.168.10.128/26 — Usable: 192.168.10.129–192.168.10.190 — Broadcast: 192.168.10.191.
- Subnet 4 — Network: 192.168.10.192/26 — Usable: 192.168.10.193–192.168.10.254 — Broadcast: 192.168.10.255.

2) For one of the subnets you created, assign valid IP addresses to 5 devices (such as a router, two laptops, a printer, and a mobile) and list which IP goes to which device.

Step 1: Select a Subnet

- Use Subnet 1: 192.168.10.0/26.
- Do not use 192.168.10.0 because it is the network address.
- Do not use 192.168.10.63 because it is the broadcast address.

Step 2: Assign Device IP Addresses

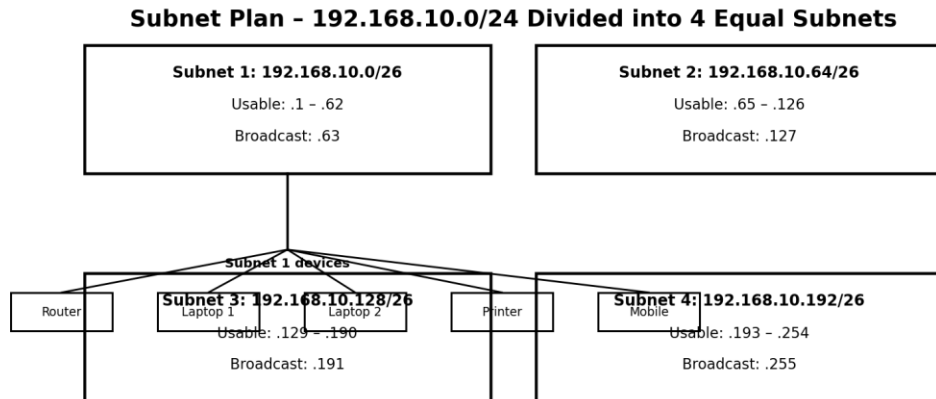
- Router: 192.168.10.1
- Laptop 1: 192.168.10.2
- Laptop 2: 192.168.10.3
- Printer: 192.168.10.4
- Mobile: 192.168.10.5
- Subnet mask for all five devices: 255.255.255.192.

IP Address Table

- Router → 192.168.10.1
- Laptop 1 → 192.168.10.2
- Laptop 2 → 192.168.10.3
- Printer → 192.168.10.4
- Mobile → 192.168.10.5

3.) Draw a simple diagram (hand-drawn or digital) showing your subnet plan: label each subnet, the range of IPs, and where each device is connected.

Subnet Plan Diagram



- The diagram below shows all four /26 subnets and the five example devices connected within Subnet 1.

Device Connections

- Router uses 192.168.10.1.
- Laptop 1 uses 192.168.10.2.
- Laptop 2 uses 192.168.10.3.
- Printer uses 192.168.10.4.
- Mobile uses 192.168.10.5.

4) Document step-by-step how you calculated the subnet mask, number of hosts per subnet, and assigned IP addresses for your plan, as if you are explaining it to a friend who is new to subnetting.

Step 1: Start with the Original Network

- The starting network is 192.168.10.0/24.
- A /24 leaves 8 host bits.

Step 2: Borrow Bits for Four Subnets

- Four subnets are required.
- 2 bits are borrowed because $2^2 = 4$.
- The prefix changes from /24 to /26.

Step 3: Find the Subnet Mask

- /26 corresponds to 255.255.255.192.
- The last octet increases in blocks of 64.

Step 4: Find the Number of Hosts

- There are 6 host bits remaining because $32 - 26 = 6$.
- $2^6 = 64$ total addresses per subnet.
- Two addresses are reserved for the network and broadcast addresses.
- Therefore, $64 - 2 = 62$ usable host addresses per subnet.

Step 5: Find Each Network and Broadcast Address

- Start at 0 and add the block size of 64: 0, 64, 128, and 192.
- The next subnet address is the previous subnet's broadcast address plus 1.
- The broadcast address is one address before the next subnet.

Step 6: Assign Device Addresses

- Choose addresses only from the usable host range.
- For Subnet 1, the usable range is 192.168.10.1 to 192.168.10.62.
- The first five usable addresses were assigned to the router, two laptops, printer, and mobile.

Final Result

- Subnet mask: 255.255.255.192 (/26).
- Number of equal subnets: 4.
- Usable hosts per subnet: 62.
- Selected device addresses: 192.168.10.1 through 192.168.10.5.